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Boarding strategies accounting for properties of the blended wing body aircraft

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ABSTRACT

Although the blended wing body aircraft has been long speculated to be the next-generation passenger aircraft, its boarding processes remains relatively unexplored. In this study, we investigate possible boarding strategies for the blended wing body aircraft. The uncertainty in the design of the blended wing body aircraft and future civil aviation industry requires that boarding strategies must be evaluated from multiple perspectives, including adaptability, practicality, boarding rate, occupancy rate, passenger compliance, etc. We propose a novel boarding strategy specifically designed for the blended wing body aircraft, which can be used for different blended wing body aircraft. Performances of boarding strategies adapted from those commonly used today are also evaluated. As a result, while the performance of the novel boarding strategy is limited by operational constraints, it demonstrates better efficiency and robustness compared to conventional boarding strategies.

1. Introduction

For passenger aircraft, boarding is on the critical path of the turnaround [1,2]. As the final step in the turnaround, boarding determines the estimated off block time of the aircraft, thereby influencing the daily operations of aircraft and airports [3–5]. A short and predictable boarding time is the key to improving the overall operation efficiency. It is estimated that for a major airline with more than 5000 flights per day (out of approximately 100000 flights taking off around the world every day), a single minute reduction in turnaround time across all flights would lead to cost savings of more than 50 million dollars per year [6]. Airports, especially busy hub airports, can benefit from streamlining the boarding process, ensuring maximum efficiency in serving aircraft at each gate. Passengers would also appreciate methods aimed at reducing waiting time at the gate and travel delays, improving their boarding experience [7]. On the other hand, accurately predicting all aircraft-related processes along specific routes is crucial for ensuring punctual operations. The time variability in arrival mainly results from that of departure, and boarding is the most unpredictable part of the process. The time required for the flight itself is generally predictable. Once the aircraft has landed, however, the uncertainty quickly increases. Intra-European flight data in 2008–2015 shows that the standard deviation for the flight phase is 5.3 min, significantly lower than that of both departure (16.6 min) and arrival (18.6 min) [8]. While other tasks during the turnaround, such as catering, unloading, and cleaning, follow predefined procedures and are largely controlled by trained personnel, boarding largely relies on passengers'

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willingness to follow instructions. Situations like late arrivals, no-shows, priority passengers, the amount of carry-on baggage, and the unwillingness to follow instructions could all significantly influence the boarding time [8].

Current studies on aircraft boarding primarily focus on minimizing the boarding time of conventional tube-and-wing passenger aircraft through specific boarding sequence designs. However, despite recent improvements, airlines do not always use efficient boarding strategies [6,9]. Simulations (e.g., [5,10–12]), analytical methods (e.g., [13]), and experiments (e.g., [14]) have been used to study the boarding problem. A comprehensive review of the optimization model for boarding has been given by Bidanda et al. [15]. Generally, the simulation-based approach has been widely applied because it is significantly easier to implement compared to experimenting with a simulated/real cabin environment.

Boarding delays are mainly caused by two types of passenger interferences, aisle and seat. Aisle interference occurs when a passenger stops in the aisle (usually resulting from stowing baggage), preventing other passengers from walking further. Seat interference occurs when a seated passenger must stand up and make way for other passengers [7]. Compared with seat interferences, aisle interferences are more difficult to reduce and usually cause more delays. To reduce the number of passenger interferences, passengers can be organized into a specific boarding sequence either before or after passengers pass through the boarding gate. However, this sorting process inherently requires cooperation from passengers and a potentially long time [16,17]. Coppens et al. [18] suggest that it is possible to reduce the influence of aisle interferences through passenger training and preparations. Tang et al. [19,20] develop a boarding model considering grouping behaviors, maximum speed of individual passengers, and the amount of baggage. Milne and Salari [21] propose a boarding strategy that assigns passengers to a specific numerical position in line based on their seat location, which is determined by the amount of baggage. Qiang et al. [22] examine the boarding strategy where passengers are not assigned to predetermined seats and can choose a preferred seat. Kierzkowski and Kisiel [23] provide an analysis of baggage stowing time from the perspectives of seat availability and occupancy of the aircraft.

In real-life boarding, boarding strategies commonly used by airlines include Random (all passengers enter the aircraft randomly), Outside-in (also known as "window-middle-aisle", passengers assigned to window seats board first, then middle seats and aisle seats), Back-to-Front (passengers assigned to the back seats board first, then middle seats and front seats) [24]. Depending on the airline and the airport, the implementation of boarding strategies can range from being highly regulated to slightly regulated. A highly regulated boarding combined with certain methods (use two boarding doors, reduce demands for catering and cleaning, etc.) could potentially reduce the turnaround time down to 25 minutes [25]. As reported by Mas et al. [26], for a single-aisle aircraft, it is estimated that Outside-in performs better than Random and Back-to-Front. However, real-life boardings are influenced by a large number of factors and boarding strategy may not function as well as expected. For example, Outside-in is relatively difficult to apply in real life because it separates small groups of passengers. An optimal boarding strategy for a single-aisle aircraft was proposed by Steffen [14,27] under an ideal scenario, which mainly focuses on minimizing aisle interferences caused by the passengers stowing baggage in the overhead compartment. It arranges all passengers in a predetermined sequence individually for maximum boarding efficiency. While this optimal boarding strategy is usually considered highly ideal and impractical, some Brazilian airports have been trying to apply this boarding strategy with assistance from floor projections [28]. Naturally, asking all passengers to follow such highly specific directions remains difficult in real life, whether they are from the airport staff or projection lamps. For example, passengers' interest to board in small groups is not respected and could lead to unwillingness to follow instructions [7].

While boarding strategies for modern aircraft have been widely studied, those for the blended wing body (BWB) aircraft have received much less attention. BWB aircraft is usually considered as the next-generation aircraft for civil aviation. It is expected to meet the growing demands of civil aviation in the future, including better fuel efficiency, lower noise, and reduced emissions [29,30]. Previous discussions of boarding strategies for the BWB aircraft are limited to simple ones such as Random and Outside-in [31]. Given that the cabin has a significantly different shape compared to modern aircraft, it is natural to assume that more boarding strategies should be available. On the other hand, although a large number of different designs exist for the BWB aircraft [32–34], its cabin design remains unclear, which has a significant influence on the final design of the aircraft. The cabin design and the BWB aircraft itself had become significantly smaller over the past 30 years, after carrying a large number of passengers (approximately 700–1000) is no longer considered as a priority for future passenger aircraft [35–37]. As another example, while most BWB designs place the cabin inside the upper half of the fuselage, the BWB-300 in [38] places the cabin at the lower half of the fuselage to ensure that the cabin will still function in a similar way to modern aircraft. However, this leaves out questions such as where the cargo bay should be and whether boarding bridges can safely approach the boarding door under the wing. Design details such as whether a boarding door on the side is possible and where boarding doors should be remain unclear. Naturally, it is necessary to explore this issue from the perspective of boarding and provide references for future studies on BWB designs.

This study comprehensively investigated the boarding strategies for future BWB aircraft. A cellular automaton boarding simulation model is developed to simulate five different boarding strategies. Four conventional boarding strategies are adapted from those commonly used today. The fifth boarding strategy is an efficient and flexible boarding strategy specifically designed for the BWB aircraft. Performances of five boarding strategies are evaluated based on the boarding time as well as the time variability. The robustness of all boarding strategies is also discussed from the perspectives of passengers and operational constraints. This study also tries to explore how cabin designs of the BWB aircraft and future airport infrastructures interact with boarding. Unlike modern aircraft, whose cabin designs are generally fixed, it is entirely possible to improve the cabin design for passengers during the conceptual design phase of future BWB aircraft.

The remainder of this paper is organized as follows. The simulation model used in this study is described in Section 2. Section 3 and Section 4 focus on the performances of all five boarding strategies. Section 5 discusses the robustness of boarding strategies. Finally, conclusions and related discussions are presented in Section 6.

2. Boarding model

2.1. Model layout

We developed a boarding model based on the cellular automaton (CA) model, which was successfully adapted to model the boarding process as well as other forms of traffic flows [39,40]. The CA model discretizes space into cells, which have two states: occupied or empty. A cell can be temporarily occupied by an agent (passenger walking on aisles), or permanently occupied by obstructions (walls, seats, etc.). Each cell is a 0.4 × 0.4 m square. Similar setups of cellular automaton boarding models have been well-tested and validated during previous studies [41–43].

The layout of the boarding model is shown in Fig. 1. We mainly referred to NASA's N2A-HWB [33] for the cabin design. Fig. 2 shows the 3D model of a N2A-HWB without engines. Unfortunately, no cabin layout information is available. Based on data from the model, the cabin is 14.924 m wide. Assuming that each bay is 3.66 m wide by default [34], a four-bay layout is the most reasonable choice. Each bay contains 13 rows of seats, and each row has six seats.

In this model, the boarding process is simulated by the movements of agents. We assumed that the model has a seat load factor (the percentage of seats that will be occupied by agents at the end of the simulation) of 1; therefore, 312 agents were used in the simulation. Further discussions on the seat load factor are covered in Section 5.3. The total number of seats was comparable to that of a modern wide-body passenger aircraft such as the Airbus A350 or Boeing B787. Only the economic class was considered in the boarding simulations. Based on previous BWB designs [45,46], business and first classes are likely located on the left-hand side of the passenger boarding bridge. We assume that business class and first class passengers board before the economy class, similar to the boarding of modern aircraft.

The presence of windows in a BWB aircraft remains uncertain because of its unique cabin shape [47]. We still use the term "window seats" to refer to the outermost seats in each bay. This does not imply the presence of physical windows adjacent to these seats. We also assumed that all bays are separated by walls. During boarding, agents cannot move between bays after entering one. Given the technical difficulties of building a wide pressurized cabin, these walls work as the necessary supporting structures [35]. They also ensure that the number of seats per row is not too high for passenger comfort.

As a time-discrete model, the simulation continues until all agents are seated, and the boarding time is the total number of timesteps required for this process. This model is based on the Mesa framework [48].

2.2. Agents' behaviors

In this model, the agents' behaviors follow a set of predefined rules: a) entering the aircraft through the passenger boarding bridge (generated at the entrance if the cell is vacant), b) moving on the main aisle to reach the corresponding bay c) move to reach the assigned seat row d) stowing their baggage, e) taking their seats. An agent will move from cell to cell unless the next cell is occupied.

2.2.1. Entering

Agents are generated at the boarding door to simulate passengers entering the aircraft. Each agent is given a specific seat and a baggage stowing time before being generated. Their order is determined by the boarding strategy. If congestion happens and the entrance is occupied, agents are added to a "waiting queue" instead of being generated, representing agents waiting on the passenger board bridge. If the entrance is no longer occupied, agents in the waiting queue are generated first.

We assume the boarding rate is 60 passengers per minute, which is significantly higher than the typical boarding rate of 14–15 passengers per minute [43,49]. The boarding rate is the arrival frequency of passengers at a single boarding door. This is different from

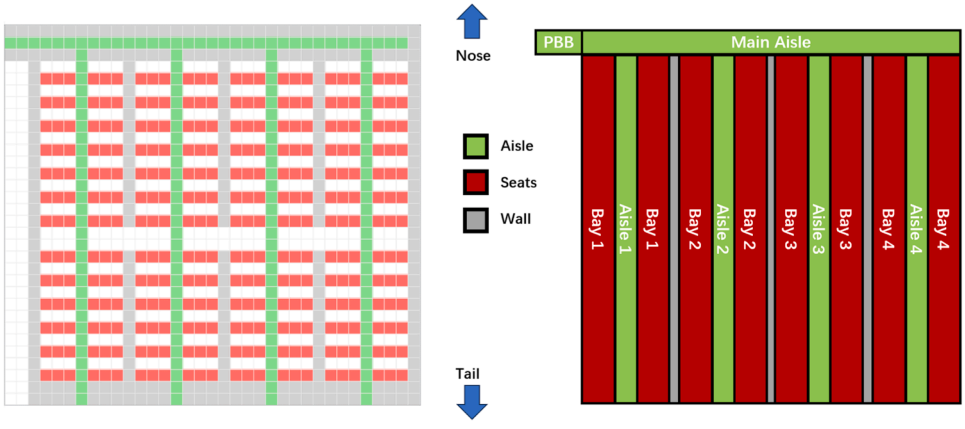


Fig. 1. Overview of the boarding model (left) and names for all structural elements (right). The 32 × 35 grid contains the BWB cabin and a small part of the passenger boarding bridge (PBB). The cabin is divided into four separate bays, each with its own aisle. The four aisles are connected to the main aisle on the top.

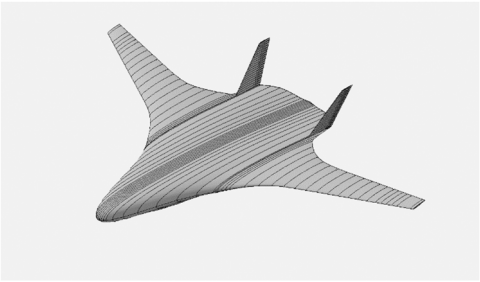


Fig. 2. Model of N2A-HWB in vsp3 format, available from VSP AirShow [44].

the rate of passengers entering the aircraft, because congestion might happen and stop passengers from entering. Currently, boardings in the standard turnaround process of Airbus A350, Boeing B787, and other similar-sized twin-aisle aircraft typically involve two boarding doors (in practice, it is also common to use only one boarding door for the economy class). While a typical modern type A boarding door defined in Federal Aviation Administration regulations is more than 42 in. wide [50] and permits a substantially higher boarding rate, increasing the boarding rate will not improve boarding efficiency because of congestion. However, the presence of multiple aisles in the BWB boarding model greatly reduces congestion at the boarding bridge, thus enabling a higher boarding rate.

On the other hand, whether future BWB aircraft can have two boarding bridges remains an open question. A second boarding door in the middle of the BWB aircraft could lead to difficulties in manufacturing. Whether it is accessible for boarding bridges is also an unsolved issue. Another option is to use the door on the far side of the BWB aircraft as a second entrance. If we separate the bays into "2 + 2" or "3 + 1", ideally most boarding strategies will still work. However, unlike modern aircraft, the lack of middle doors means that we cannot separate one aisle into two. Even if we allow passengers to board from two doors, boarding speed will still be limited by aisle interferences on four aisles. From this perspective, boarding from two doors is not very different from one door with a high boarding rate. Reaching the far side of the aircraft could be another issue. Naturally, using a second boarding bridge could be very difficult. If shuttle buses are used for both doors, it could be challenging to rearrange the order of passengers after they exit from the bus.

In this study, we assume that only one boarding bridge is available and the boarding rate is 60 passengers per minute unless otherwise specified. Presumably, the boarding door is wider than the modern type A boarding door. We chose this high boarding rate to better evaluate the effectiveness of different boarding strategies. Performances of efficient boarding strategies can be limited by boarding rates. A more detailed discussion on boarding rate and boarding door is covered in Section 5.1.

2.2.2. Moving

Agents move to their assigned seats from cell to cell. During each timestep, the model uses a sequential update (agents are updated based on the order they are generated) [51]. Notably, because other pedestrian simulations frequently use parallel updates, an agent must wait for a timestep before moving to another cell that was occupied by another agent, regardless of whether the latter had already left the cell within the same timestep. Considering that passengers typically carry baggage and stop frequently on the aircraft, we use a sequential update to ensure pedestrians can move slowly but seamlessly, ignoring the aforementioned movement delay. Each timestep is defined as 0.5 s. The walking speed is set as 0.8 m/s. [42,43].

2.2.3. Baggage stowing

As in real-life boarding, different agents require different amounts of time for baggage stowing. Field data from German airports indicates that the distribution of baggage stowing time (t_b) can be approximated by Weibull distribution [52]:

$$t_b(x, \alpha, \beta) = round(1 - e^{-\left(\frac{x}{\beta}\right)^{\alpha}}), \tag{1}$$

where $\alpha = 1.7,\ \beta = 16.0 seconds$ [52]. *round* means that the number is rounded to the nearest integer for timesteps defined in the CA model. t_b is randomly determined before the agent enters the aircraft. On average, an agent requires 14 seconds for baggage stowing. Most agents require 5–25 seconds, although extreme cases with 35 seconds may be generated. This can be attributed to reasons such as the age of passengers, the type and quantity of luggage, the existing luggage in the overhead compartment, etc. [18,53] The minimum baggage stowing time is 0 seconds, meaning that the agent has no carry-on baggage or it is small enough to stow it under the seat after the agent is seated.

Baggage stowing causes aisle interference where the leading agent performing baggage stowing stops the entire queue behind it. This is usually the most important reason for boarding delays. Boarding strategies are generally designed to minimize the number of aisle interferences and increase the number of agents that can perform baggage stowing simultaneously.

2.2.4. Seating

After stowing their baggage, agents will move to their seats and sit down. If the agent is seated on an aisle seat or a middle seat, another agent might need to go over this seat to reach its seat. In this case, the first agent must stand up and move to the aisle in order to

make way for the second agent. This is usually referred to as a seat interference, which is another reason for boarding delays. In our model, agents involved in seat interferences are temporarily given the highest update priority to ensure that it can be resolved.

3. Conventional boarding strategies

In this section, we examine four boarding strategies adapted from those commonly used nowadays. Each bay within the BWB aircraft is generally similar to the cabin of a single-aisle aircraft, therefore we focus on those boarding strategies designed for single-aisle aircraft.

3.1. Simple boarding strategies

Three simple boarding strategies are adapted for the BWB aircraft.

Random: The boarding sequence is completely random. This is usually achieved by announcing that boarding has started and all passengers should come to the boarding gate.

Back-to-Front: Agents are organized into four boarding groups based on their seat positions in the longitudinal direction, regardless of which bay they belong to. Agents assigned to seats at the back area of each bay board first, followed by agents assigned to the middle areas and front area. The boarding sequence within each group is randomized. *Back-to-Front* is shown on the left side of Fig. 3. This can be achieved by printing the boarding group number on the boarding pass, then announcing which boarding group should come to the boarding gate.

Outside-in: Agents are organized into three boarding groups. The first group comprises 104 agents assigned to window seats, and the second and third groups comprise agents assigned to middle seats and aisle seats, respectively. The boarding sequence within each group is randomized. *Outside-in* is shown on the right side of Fig. 3. Similar to *Back-to-Front*, this could also be easily achieved by printing the boarding group number on the boarding pass.

The results of the three boarding strategies are shown in Fig. 4. 10000 simulations are performed for each boarding strategy. *Outside-in* (533.61 seconds) performs best in terms of average boarding time, followed by *Random* (608.26 seconds) and *Back-to-Front* (710.22 seconds).

During the boarding process, congestion near the passenger boarding bridge will stop the generation of new agents. Similar situations can be observed in real-life boarding, where passengers are constrained to wait at the end of a boarding bridge.

3.2. Double outside-in

While *Outside-in* is more effective than the other two boarding strategies, it is possible to make further improvements without significantly increasing its complexity. Now we introduce another boarding strategy, *Double Outside-in*:

Double Outside-in: Each boarding group of *Outside-in* is further divided into two groups based on their seat positions in the longitudinal direction. This boarding strategy was introduced in [43], though we took the liberty to name it as *Double Outside-in*. Agents assigned to the rear window seats board first, followed by agents assigned to the front window seats, the rear middle seats, the front middle seats, the rear aisle seats, and the front aisle seats. *Double Outside-in* is shown in Fig. 5. This boarding strategy essentially increases the number of boarding groups to six, therefore it can be achieved in a similar way to *Back-to-Front* and *Outside-in*.

The results of four conventional boarding strategies are shown in Fig. 6. *Double Outside-in* clearly outperforms the conventional boarding strategies in terms of boarding time, while the time variability is generally the same as *Outside-in*.

4. N-Aisles Steffen

In this section, we introduce a new boarding strategy named *N-Aisles Steffen (N-Steffen)*. *N-Steffen* is an effective boarding strategy designed to minimize boarding time and reduce time variability, while remaining reasonably flexible for different BWB cabins.

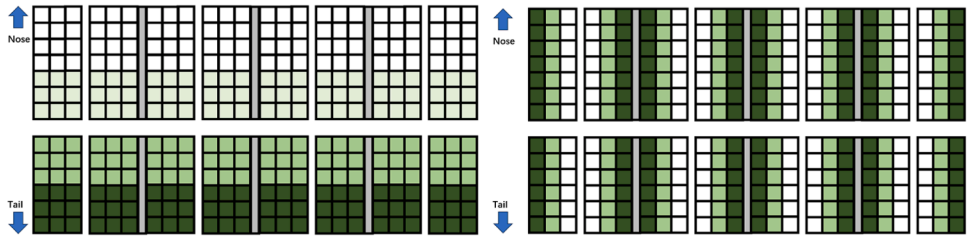


Fig. 3. *Back-to-Front* (Left) and *Outside-in* (Right) boarding strategy. Agents are organized into four and three boarding groups, respectively. Agents assigned to darker seats board first.

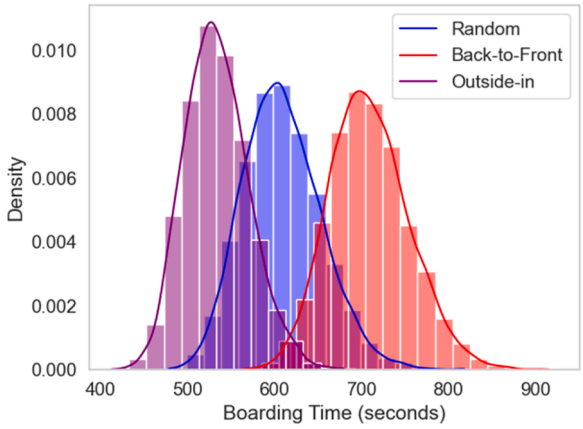


Fig. 4. Boarding time of three conventional boarding strategies. Both Gaussian kernel density distributions and histograms are plotted.

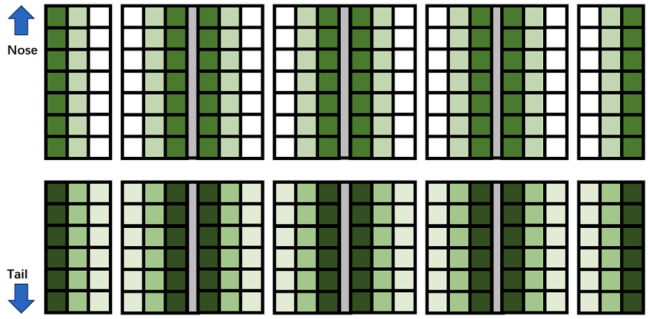


Fig. 5. *Double Outside-in* boarding strategy. Agents are organized into six boarding groups. Agents assigned to darker seats board first.

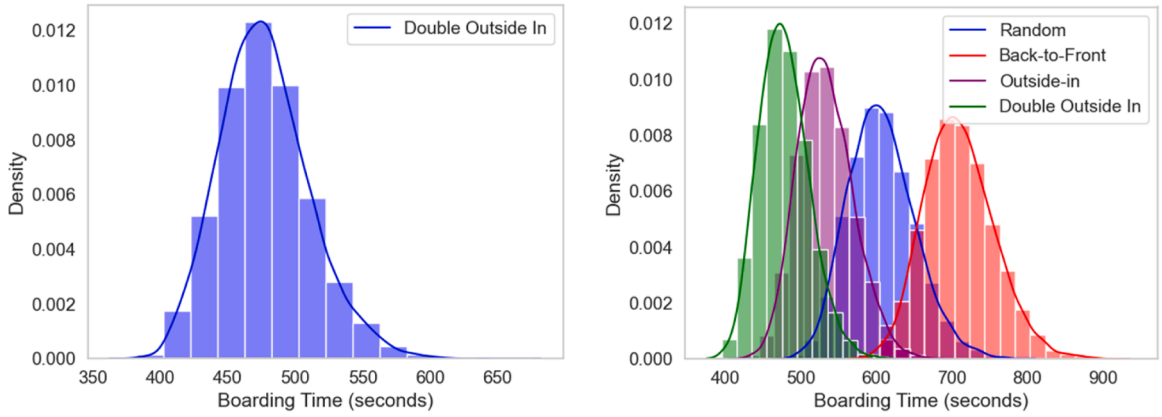


Fig. 6. Left: Boarding time of *Double Outside-in*; Right: Boarding time of four conventional boarding strategies.

4.1. Mini-groups

Steffen indicated that for a single-aisle aircraft, two passengers boarding together should have a seat difference of 12 or two rows (if every row has 6 seats) to reduce the delay caused by baggage [54]. Given that each bay in our model is similar to that of a single-aisle aircraft, we can assume that this conclusion is still applicable. The boarding problem was simplified by organizing 312 passengers into 48 mini-groups, as shown in Fig. 7. Within a mini-group, agents board from back to front. If we only consider the boarding of bay 1, Steffen's boarding strategy can be implemented by organizing the mini-groups in the following order (head to tail): 12–7–6–1–11–8–5–2–10–9–4–3.

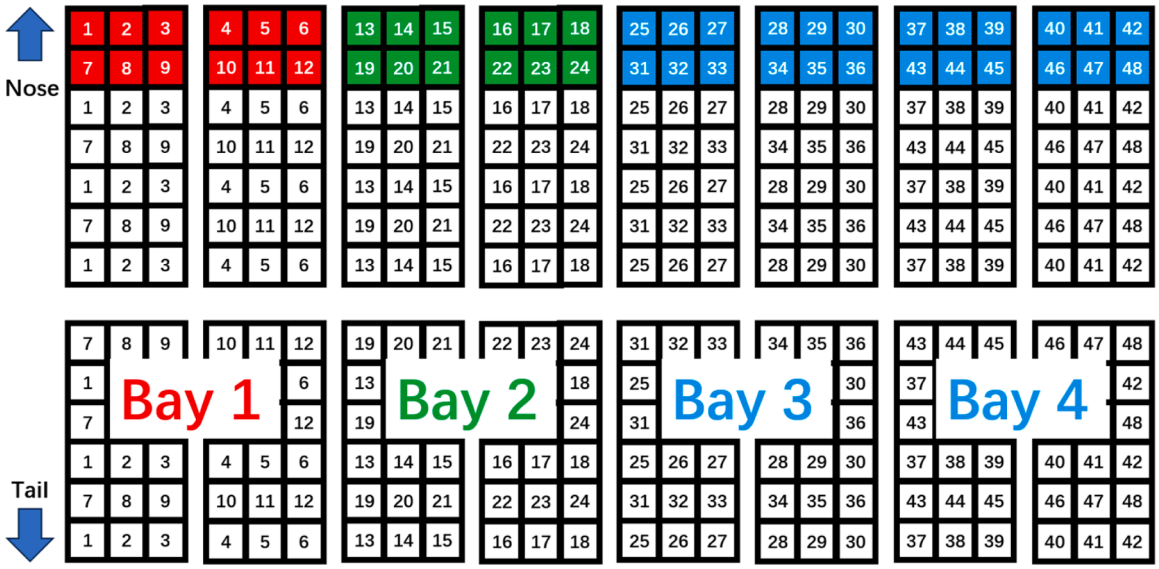


Fig. 7. 312 agents are organized into 48 mini-groups. Numbers on the seats indicate the mini-group to which the corresponding agent is assigned. We may focus on the 48 agents in the first two rows (marked in red, green, and blue), each representing a mini-group. Both Bay 3 and Bay 4 are marked in blue because agents belonging to these two bays have a smaller influence on the boarding time, as explained in Sec. 3.3.2.

4.2. *N-Aisles Steffen*

Compared with conventional tube-and-wing aircraft, additional aisles on BWB passenger aircraft significantly influence the boarding process. Board sequentially for each bay would result in a long boarding time, because the additional aisles are not used. We now introduce this boarding strategy as the starting point of our discussion.

Sequential Steffen (S-Steffen): Each bay is filled sequentially with the classic *Steffen Perfect* [6]. In *Steffen Perfect*, agents are first assigned to windows seat boards in a back-to-front order while skipping one row. This is the optimal boarding strategy for a single bay in our simulation model (or a single-aisle aircraft), which may reduce boarding time by more than a factor of four under ideal scenarios [6]. Bay 4 is boarded first, followed by Bay 3, Bay 2, and Bay 1.

Naturally, a modified version of *Sequential Steffen* that allows all four bays being filled simultaneously should be significantly more efficient. As shown in Fig. 8, we assume that the following eight mini-groups should board first: 7, 12, 29, 24, 31, 36, 43, and 48. These eight mini-groups were the first two mini-groups to board each bay when *S-Steffen* was applied. Minimizing boarding time means that we must reduce the number of agents influenced by congestion; therefore, congestion at the entrances of Bay 3 and Bay 4 is acceptable provided that the queue is not exceedingly long. However, congestion at the entrances of bays 2 and 1 must be prevented.

A combined group is defined as two mini-groups boarding together. Combined groups are classified into two types based on their influences on congestion:

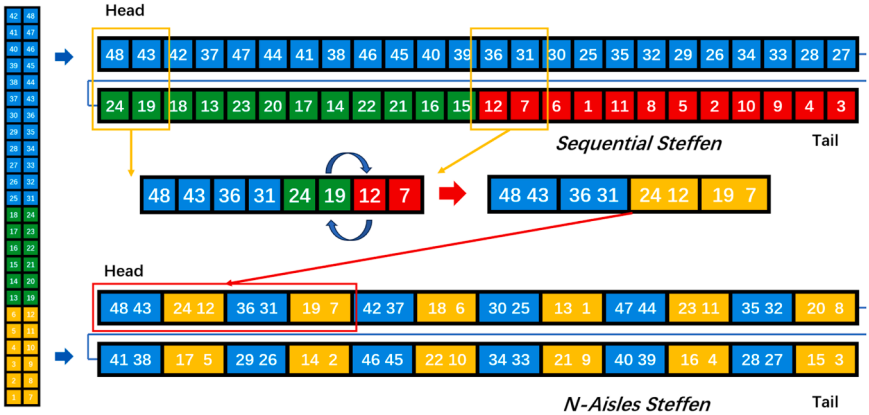


Fig. 8. Modification of *Sequential Steffen* into *N-Aisles Steffen*. This strategy ensures that for bays 1 (red) and 2 (green), only one combined group will attempt to enter the corresponding aisle simultaneously. Type A combined groups are yellow and type B combined groups are blue. Mini-groups or combined groups on the left board first.

Type A: Mini-groups belonged to bay 1 and bay 2. This type of mini-group has a significant impact on the boarding process. One type A combined group is made of one mini-group from bay 1 and another one from bay 2, therefore one type A combined group can minimize the congestion at the entrances of bay 1 and bay 2. However, if two of them board together, congestion will still happen

Type B: Mini-groups belonged to other bays. This type of mini-group has a limited impact on the boarding process.

We now introduce the boarding strategy *N-Aisles Steffen (N-Steffen)*.

N-Steffen: No two type A combined groups board together. Two type A mini-groups are always separated by at least one type B combined group to reduce congestion at the entrance. Positions of two type A or type B combined groups are interchangeable.

As an example, the boarding order of the first eight mini-groups can be 48–43–24–12–36–31–19–7. By applying a similar order to the other 40 mini-groups, we created an *N-Steffen* boarding strategy. This boarding strategy ensures that two mini-groups will not attempt to enter Bay 2 or Bay 1 simultaneously. This unique advantage is shown in Fig. 9.

As shown in Fig. 10 and Table 1, the simulation results suggest that *N-Steffen* offers a higher boarding speed than other boarding strategies, including *Double Outside-in*. The average boarding time decreases to 335.43 seconds, with a standard deviation of 20.17 seconds. *N-Steffen* not only offers faster boarding, but a more stable, predictable boarding as well. On the other hand, also shows significant improvements compared to other conventional boarding strategies.

4.3. Flexibility of N-Aisles Steffen

The sequence of combined groups in *N-Aisles Steffen* is not fixed. For example, switching the order of two type A or type B combined groups does not significantly affect its performance. If we switch the position of 48–43 with 36–31 in Fig. 8 and proceed to do the same for all the other type A combined groups, the boarding time remains generally the same. Naturally, to further reduce congestion, a type B combined group may also be made of one mini-group from bay 3 and another one from bay 4. However, the influence on boarding time is relatively small (5.36 seconds faster). It is also possible to adapt it for the cabin where not all bays have an equal number of seats, because adding or removing mini-groups or combined groups is acceptable [55].

Because type B combined groups can include as many bays as required, *N-Steffen* can also be extended to BWB cabins with more bays and aisles. Fig. 11 shows that when an extra aisle is added, *N-Steffen* remains a highly effective boarding strategy. Five boarding strategies still perform in a similar way to our previous discussions. Note that while *N-Steffen* is designed to be adaptable, it is not a conclusive, optimal boarding strategy for all BWB cabins. A large number of possible BWB cabin designs have been developed, and some of them could have vastly different boarding processes. For example, if a rear ramp is added so passengers can board from the back, it would completely change how boarding works.

If there are only three aisles, for every two type A combined groups there is only one type B combined group. In this situation, congestion can still be largely reduced by applying a similar boarding strategy [55]. A twin-aisle blended wing body aircraft design is rare because the cabin is simply not wide enough, although *N-Steffen* could still be applied without type B combined groups

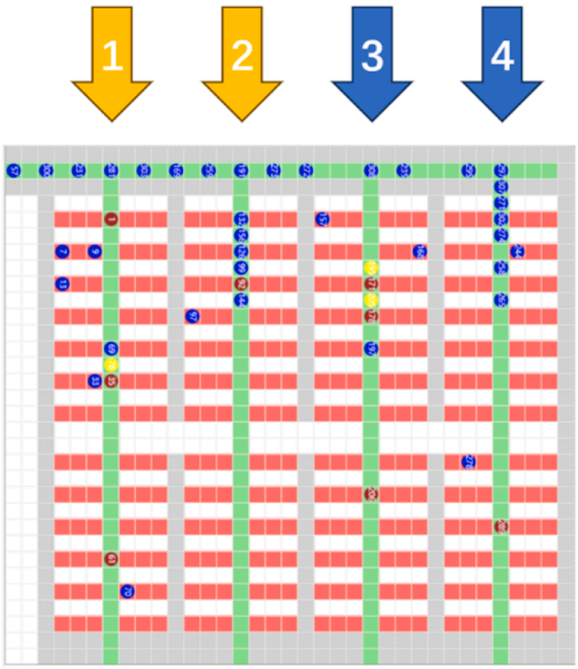


Fig. 9. Entrances to the four bays are critical to the boarding process. When *N-Steffen* is applied, congestion at entrances to bays 1 and 2 (yellow arrows) is mitigated, whereas congestion at the entrances to bay 3 and bay 4 is acceptable and is less likely to significantly influence the boarding time.

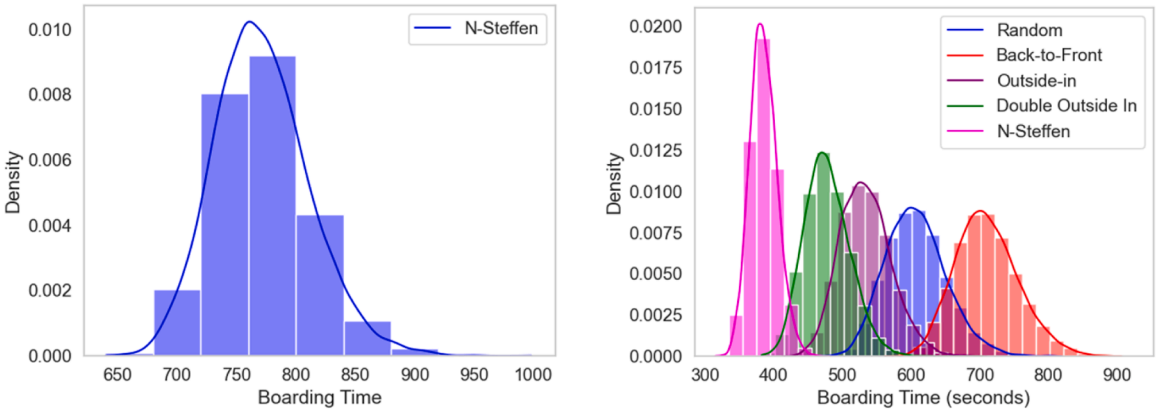


Fig. 10. Left: Boarding time of *N-Steffen*; Right: Boarding time of all five boarding strategies. Both Gaussian kernel density distribution and histograms are plotted.

Table 1
Simulation results of all five boarding strategies. Boarding time results are the average of 10000 simulations. Time variability is the standard deviation of the simulation results.

Strategy	*Random*	*Back-to-Front*	*Outside-in*	*Double Outside-in*	*N-Steffen*
Boarding Time (seconds)	608.25	710.22	533.61	478.18	335.43
Time Variability (seconds)	44.84	46.06	37.18	32.87	20.19

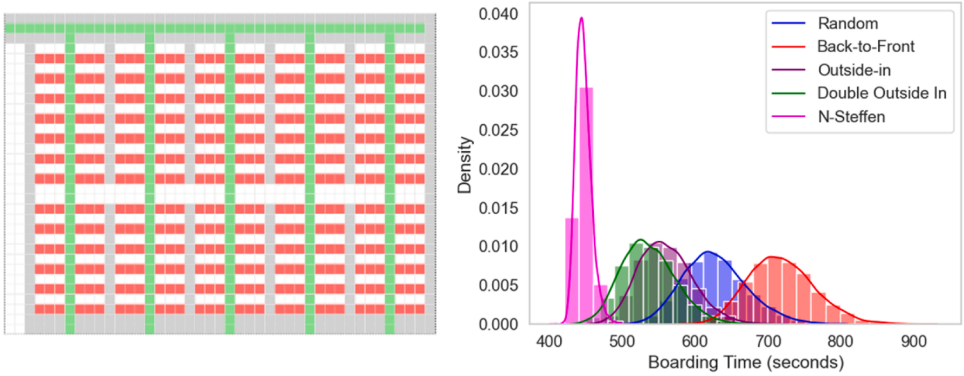


Fig. 11. Left: Adding an extra aisle in the boarding model; Right: Boarding time of all five boarding strategies when there are five aisles.

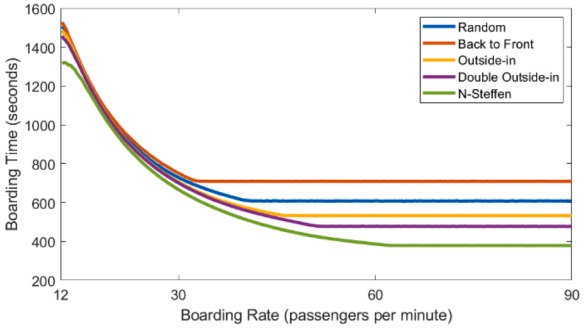


Fig. 12. How the boarding times of five boarding strategies change when the boarding rate gradually increases.

(95.93 seconds faster than *Random*). For a single-aisle scenario, *N-Steffen* simply devolves into the classic *Steffen Perfect* [6].

5. Robustness of boarding strategies

Performances of boarding strategies are influenced by a large number of factors. To test the robustness of boarding strategies, we further explore the boarding process of the BWB aircraft from four additional perspectives: boarding rates, amount of baggage, seat load factor, and compliance rate.

5.1. Boarding rates

Previous simulations assume that the boarding rate is 60 passengers per minute per boarding door. In this section, we explore the impact of boarding rates on boarding strategies' performances. For simplicity, we assume that agent generation is randomized in this section, i.e., a boarding rate is 40 passengers per minute means that there is a 40/120 chance every timestep for an agent to be generated at the entrance.

Naturally, the performances of boarding strategies are limited by boarding rates. As shown in Fig. 12, less efficient boarding strategies like *Random* and *Back-to-Front* only require a boarding rate of 30–40 passengers per minute to reach maximum performance, while N-Steffen requires more than 60 passengers per minute. However, *N-Steffen* does still have a small advantage even when the boarding rate is not high enough. On the other hand, *Double Outside-in* does not show an advantage over *Outside-in* when the boarding rate is lower than 45 passengers per minute. This indicates the limitations of *Double Outside-in*. If the boarding rate is less than 30 passengers per minute, the choice of boarding strategy does not have meaningful differences.

5.2. Amount of baggage

As the most important reason for boarding delays, baggage stowing time may vary significantly depending on the flight route. For example, the baggage stowing time we used for previous simulations is based on flights in Europe, where domestic short-haul flights are relatively common. However, in East Asian countries, domestic short-haul flights are largely substituted by trains. The baggage stowing time in East Asian airports could be higher because passengers may be carrying more baggage for longer travel distances. It is reasonable to assume that future BWB aircraft will fly on different routes where the amount of baggage can be significantly different.

The results are shown in Fig. 13. β is the constant in Eq. [1]. Different boarding strategies have different turning points where the amount of baggage begins to significantly increase boarding time. These turning points indicate that congestion begins to largely reduce the number of agents that can stow their baggage simultaneously. For *N-Steffen*, the turning point is approximately 16 seconds, which is the default value we used. Turning points for less efficient boarding strategies have lower values. Increasing the amount of baggage beyond the turning points increases boarding time linearly.

Additionally, we consider an ideal boarding scenario where passengers are discouraged from bringing carry-on baggage with them. Instead, they only have a small bag that can be put under the seat or no bags. In this case, the baggage stowing time is 2 seconds for all agents. This could also be interpreted as a classroom or a theater, given the similarity between them and a BWB cabin.

As shown in Fig. 14, if the baggage stowing time is always 2 seconds, the "boarding" process can be significantly shortened. The removal of the uncertain, time-consuming baggage stowing process has a significant impact on the boarding process. Differences in the performances of five boarding strategies are essentially removed. Boarding times are significantly shortened, and time variabilities are very low.

5.3. Seat load factor

In previous simulations, we assumed that the seat load factor is always 1. Naturally, this is not always true in real-life boarding, especially during the recent pandemic. It is necessary to explore how boarding strategies perform when the seat load factor changes.

Fig. 15 indicates that the boarding time generally increases linearly with the seat load factor (except for *Back-to-Front*), although the rates are different for different boarding strategies. If the seat load factor is lower than 0.5, the influence of boarding strategy is highly limited. The boarding time of *Back-to-Front* quickly increases after the seat load factor reaches 0.5. Similarly, this turning point also indicates the number of agents that can stow their baggage simultaneously is reduced by congestion. This effect is especially obvious for *Back-to-Front*, because one aisle interference could potentially hold up a large boarding group. Performance differences only become significant when the seat load factor is larger than 0.7.

5.4. Compliance rate

In real-life boarding, it is not uncommon that not all passengers follow the instructions required by a boarding strategy. Reasons include but are not limited to late arrivals, no-shows, groups traveling together, issues with document checks, passengers with disabilities, or simply choosing not to follow instructions. This effect is especially important for *Outside-in*, *Double Outside-in* and *N-Steffen*, because they separate agents that sit together. Therefore, in this section, we investigate how different boarding strategies perform when they cannot be perfectly implemented.

Compliance rate (R_C) is defined as the percentage of agents that are in their original correct positions when the boarding queue is generated. If the compliance rate is lower than 100 %, then some agents ($round((1 - R_C) \times 312)$, *round* means the number is rounded

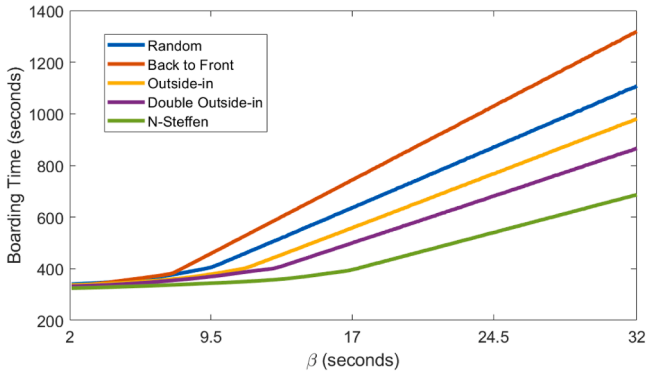


Fig. 13. How the boarding times of five boarding strategies change when the amount of baggage gradually increases.

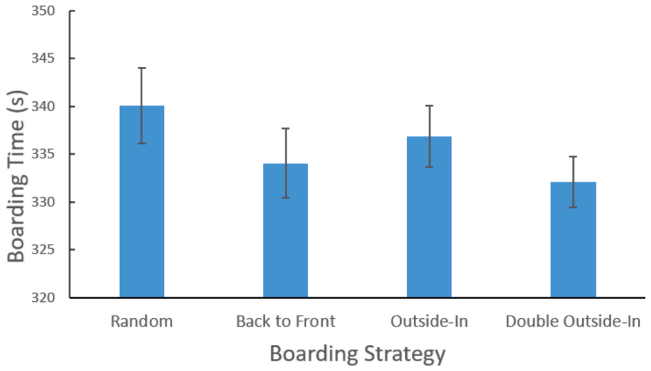


Fig. 14. Performances of four boarding strategies when the baggage stowing time is 2 seconds for all agents. *N-Steffen* ensures that no congestion happens, therefore the boarding time is always 324.5 seconds.

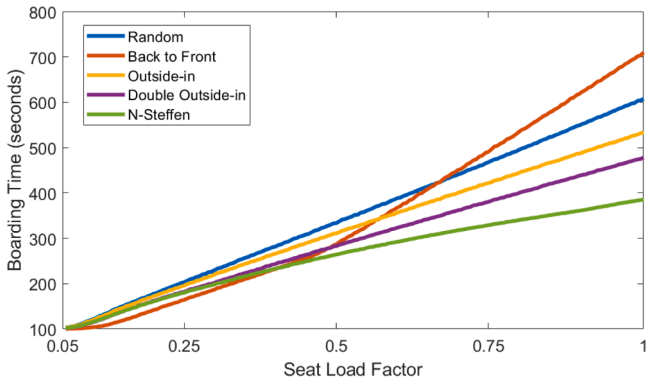


Fig. 15. How the boarding times of five boarding strategies change when the seat load factor gradually increases.

down to the nearest integer) are removed from their original positions before adding back to the boarding queue randomly. This mechanism is designed to explore the relationship between compliance and boarding strategy, instead of focusing only on grouping behaviors. Naturally, *Random* is not affected by the compliance rate.

As shown in Fig. 16, when the compliance rate gradually decreases from 100 % to 0 % (*Random*), differences in performances are gradually becoming smaller. The differences are relatively small when the compliance rate is less than 50 %. While *N-Steffen* does require a significant amount of coordination, it still performs better than other boarding strategies at the same level of compliance rate. An 80 %–95 % level compliance rate, or 15–60 agents not in their original positions, does not significantly influence the performance of *Double Outside-in*. Nevertheless, the influence on *N-Steffen* remains more significant.

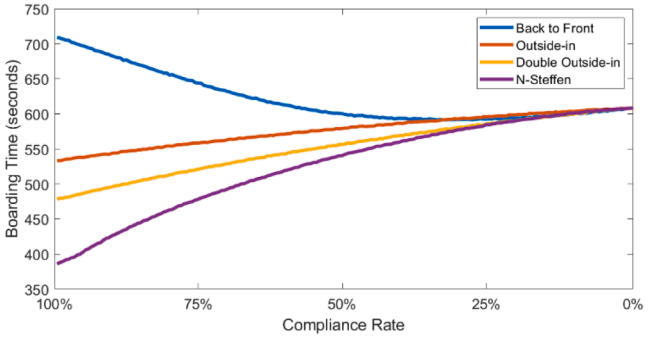


Fig. 16. How the boarding times of four boarding strategies change when the compliance rate gradually decreases.

6. Conclusion and discussion

To improve the boarding efficiency of future BWB aircraft, a short and predictable boarding process is necessary. Efficient boarding strategies can shorten the boarding time as well as reduce the time variability of boarding. However, because of the unique layout of the BWB cabin, it is also important to consider boarding during the conceptual design phase. Aircraft manufacturers must not ignore the requirements of efficient boarding to ensure a smooth boarding process. Compared to conventional tube-and-wing aircraft, the BWB cabin allows highly efficient boarding when using boarding strategies such as *Double Outside-in* and *N-Steffen*. *Double Outside-in* is a simple, practical boarding strategy that performs relatively well, while *N-Steffen* is more ideal and more efficient. Compared to *Double Outside-in* and *Outside-in*, applying *N-Steffen* in real-life boarding requires substantial on-site coordination. Nevertheless, *N-Steffen* does not require all passengers to follow directions. Even if 10 % of the passengers do not follow directions, it is estimated that *N-Steffen* still performs relatively well.

Because there are numerous different cabin designs for the BWB aircraft, *N-Steffen* is designed to be as adaptable as possible. *N-Steffen* can be adapted for different numbers of aisles or seats. Nevertheless, it is not a "one-size-fits-all" boarding strategy. We can easily imagine a scenario where passengers board the BWB aircraft from a large rear ramp, similar to how modern cargo aircraft work. In this case, *N-Steffen*, or any other boarding strategies we know today will certainly not function. If a second boarding door on the other side of the boarding bridge can be used simultaneously (assuming that using a second boarding door on the same side is not possible), presumably accessed by shuttle buses and boarding stairs, *N-Steffen* will work in a limited part of the cabin. Based on the simulation results, unlike modern twin-aisle aircraft, BWB aircraft only require one passenger boarding bridge instead of two. Because of the existence of multiple aisles, congestion at the boarding bridge is significantly mitigated.

The robustness of boarding strategies is tested with four different perspectives based on real-life boarding. Boarding rate results show that more efficient boarding strategies require a higher number of people entering the aircraft per minute. The amount of baggage passengers carry with them is the most important factor for boarding delays. If the amount of baggage can be reduced, boarding time will be shorter and more predictable. The seat load factor does not significantly influence the efficiencies of boarding strategies after it reaches a realistic level (0.7–1). A low seat load factor, however, essentially removes the differences in performance. Finally, the compliance rate results suggest that even when not all passengers choose to follow the directions of boarding strategies, performances of *Double Outside-in* and *N-Steffen* are not significantly influenced.

A critical issue for implementing more efficient boarding strategies is the boarding rate. Based on the simulation results, unlike modern twin-aisle aircraft, BWB aircraft only require one passenger boarding bridge instead of two. During the boarding of modern passenger aircraft, the boarding rate is not a bottleneck because of congestion. Because of the existence of multiple aisles on the BWB aircraft, congestion at the boarding bridge is significantly mitigated. Increasing the boarding rate to 45 passengers per minute, if possible, will improve the boarding efficiency for efficient boarding strategies such as *Double Outside-in* and *N-Steffen*. This will present new opportunities and challenges for the future civil aviation industry. Updating safety standards for boarding bridges may be necessary, and ticket checking at the boarding gate should be more automated to allow a higher boarding rate. If we only consider widely used modern technologies, airports need to adopt a ticket-checking system more similar to the urban metro system or railway system. With a proper combination of automated ticket-checking systems and ticket inspectors, it might be possible to reach a boarding rate of 60 or even 90 passengers per minute. It is also possible that future boarding passes contain RFID labels that allow passengers to simply walk through the boarding gate, similar to retailers using RFID labels for self-checkout. A wider boarding door should also be designed during the development of the BWB aircraft to accommodate the increased passenger flow.

As with modern passenger aircraft, the amount of baggage has a significant influence on boarding efficiency. It is not unimaginable that future passengers are discouraged from bringing in too much baggage with them. The large internal space of the BWB aircraft may make check-in baggage cheaper and more accessible, meaning that fewer passengers will carry big backpacks onboard. Onboard entertainment devices may become more powerful and can access the internet through low earth orbit satellite constellations, reducing the need for bringing a laptop or a tablet.

The compliance rate must remain above a certain threshold to ensure boarding strategies function normally. Even though the compliance rate needs to be as low as 50 % to make *Double Outside-in* and *N-Steffen* unfavorable, a threshold of 75 % or 80 % might be a

more reasonable option. While it is difficult to predict the compliance rate for all boarding, certain special cases might be worth considering. For example, a popular flight route to a holiday destination during public holidays is highly likely filled with passengers traveling in small groups that do not want to board separately. We may safely assume that the compliance rate will be lower than usual. Using *Outside-in*, *Double-Outside-in*, and *N-Steffen* is likely to make the compliance rate even lower, because these passengers likely have the seats on the same row. In such a situation, airlines may simply ask passengers to board randomly.

As mentioned above, it is difficult to predict boarding-related details at that time. For example, virtual reality and augmented reality technologies may become widely adopted, so that boarding can be guided in a virtual way such as a head-up display. An extremely important application for this is that after entering the aircraft, passengers must find out which aisle they should go to. Currently, this is usually done by an air attendant telling passengers the directions after looking at their boarding passes. However, this may not be efficient enough for the BWB aircraft, given that the boarding rate will be significantly higher. Simply printing the bay number on the boarding pass could be a "low-tech" alternative, although this may not be very efficient for such a high boarding rate. One passenger not understanding the situation could potentially significantly slow down the boarding.

While the BWB aircraft has been predicted to revolutionize modern civil aviation industry for many years, this massive potential paradigm shift remains as challenging as ever. Our best hope is that our research could reduce the impact of this paradigm shift by providing new perspectives for aircraft designers, airlines, and airports. For example, if boarding speed can be significantly increased compared to modern aircraft, airlines and airports will have more incentives to push for this massive paradigm shift, instead of pushing against it. On the other hand, even if we have a fully functional BWB aircraft today, it will still require completely new standards for operation and maintenance. We cannot ignore the fact that many practices and standards for modern civil aviation come from terrible air accidents in the past. Going through this path again will certainly have a massive negative impact on the reputation of BWB aircraft, while it is already difficult enough to convince passengers to accept this completely new type of aircraft. Instead of focusing completely on the BWB passenger aircraft, a more reasonable approach could be developing BWB cargo airplanes, ideally piloted by advanced artificial intelligence. We hope that this approach will gradually improve BWB aircraft's public image and accumulate the necessary experience for manufacturers and maintenance personnels.

CRediT authorship contribution statement

Yuming Dong: Writing – original draft, Software, Methodology, Conceptualization. **Xiaolu Jia:** Writing – review & editing, Methodology. **Daichi Yanagisawa:** Writing – review & editing, Supervision, Resources, Conceptualization. **Katsuhiro Nishinari:** Supervision, Resources.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT-4o in order to polish the writing. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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